



1. Description

1.1. Project

Project Name	XDIC TRIM JIG
Board Name	NUCLEO-F446RE
Generated with:	STM32CubeMX 6.12.0
Date	11/12/2024

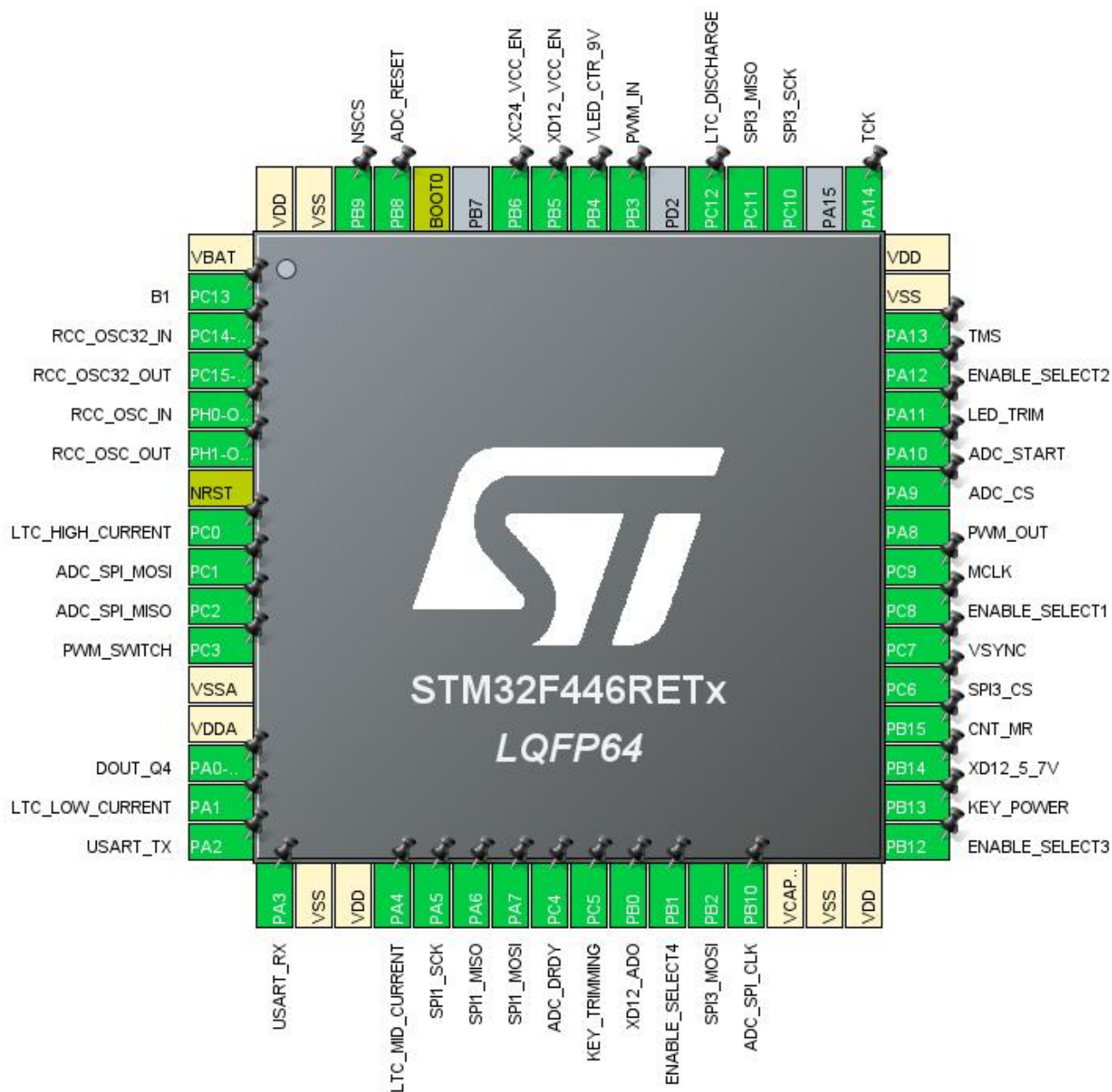
1.2. MCU

MCU Series	STM32F4
MCU Line	STM32F446
MCU name	STM32F446RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	Arm Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
2	PC13 *	I/O	GPIO_Input	B1
3	PC14-OSC32_IN	I/O	RCC_OSC32_IN	
4	PC15-OSC32_OUT	I/O	RCC_OSC32_OUT	
5	PH0-OSC_IN	I/O	RCC_OSC_IN	
6	PH1-OSC_OUT	I/O	RCC_OSC_OUT	
7	NRST	Reset		
8	PC0 *	I/O	GPIO_Output	LTC_HIGH_CURRENT
9	PC1	I/O	SPI2_MOSI	ADC_SPI_MOSI
10	PC2	I/O	SPI2_MISO	ADC_SPI_MISO
11	PC3 *	I/O	GPIO_Output	PWM_SWITCH
12	VSSA	Power		
13	VDDA	Power		
14	PA0-WKUP	I/O	TIM5_CH1	DOUT_Q4
15	PA1 *	I/O	GPIO_Output	LTC_LOW_CURRENT
16	PA2	I/O	USART2_TX	USART_TX
17	PA3	I/O	USART2_RX	USART_RX
18	VSS	Power		
19	VDD	Power		
20	PA4 *	I/O	GPIO_Output	LTC_MID_CURRENT
21	PA5	I/O	SPI1_SCK	
22	PA6	I/O	SPI1_MISO	
23	PA7	I/O	SPI1_MOSI	
24	PC4	I/O	GPIO_EXTI4	ADC_DRDY
25	PC5 *	I/O	GPIO_Input	KEY_TRIMMING
26	PB0	I/O	ADC1_IN8	XD12_ADO
27	PB1 *	I/O	GPIO_Output	ENABLE_SELECT4
28	PB2	I/O	SPI3_MOSI	
29	PB10	I/O	SPI2_SCK	ADC_SPI_CLK
30	VCAP_1	Power		
31	VSS	Power		
32	VDD	Power		
33	PB12 *	I/O	GPIO_Output	ENABLE_SELECT3
34	PB13 *	I/O	GPIO_Input	KEY_POWER
35	PB14 *	I/O	GPIO_Output	XD12_5_7V
36	PB15 *	I/O	GPIO_Output	CNT_MR

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
37	PC6 *	I/O	GPIO_Output	SPI3_CS
38	PC7	I/O	TIM8_CH2	VSYNC
39	PC8 *	I/O	GPIO_Output	ENABLE_SELECT1
40	PC9	I/O	RCC_MCO_2	MCLK
41	PA8	I/O	TIM1_CH1	PWM_OUT
42	PA9 *	I/O	GPIO_Output	ADC_CS
43	PA10 *	I/O	GPIO_Output	ADC_START
44	PA11 *	I/O	GPIO_Output	LED_TRIM
45	PA12 *	I/O	GPIO_Output	ENABLE_SELECT2
46	PA13	I/O	SYS_JTMS-SWDIO	TMS
47	VSS	Power		
48	VDD	Power		
49	PA14	I/O	SYS_JTCK-SWCLK	TCK
51	PC10	I/O	SPI3_SCK	
52	PC11	I/O	SPI3_MISO	
53	PC12 *	I/O	GPIO_Output	LTC_DISCHARGE
55	PB3	I/O	TIM2_CH2	PWM_IN
56	PB4 *	I/O	GPIO_Output	VLED_CTR_9V
57	PB5 *	I/O	GPIO_Output	XD12_VCC_EN
58	PB6 *	I/O	GPIO_Output	XC24_VCC_EN
60	BOOT0	Boot		
61	PB8 *	I/O	GPIO_Output	ADC_RESET
62	PB9 *	I/O	GPIO_Output	NSCS
63	VSS	Power		
64	VDD	Power		

* The pin is affected with an I/O function



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32F4
Line	STM32F446
MCU	STM32F446RETx
Datasheet	DS10693 Rev6

1.2. Parameter Selection

Temperature	25
Vdd	3.3

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

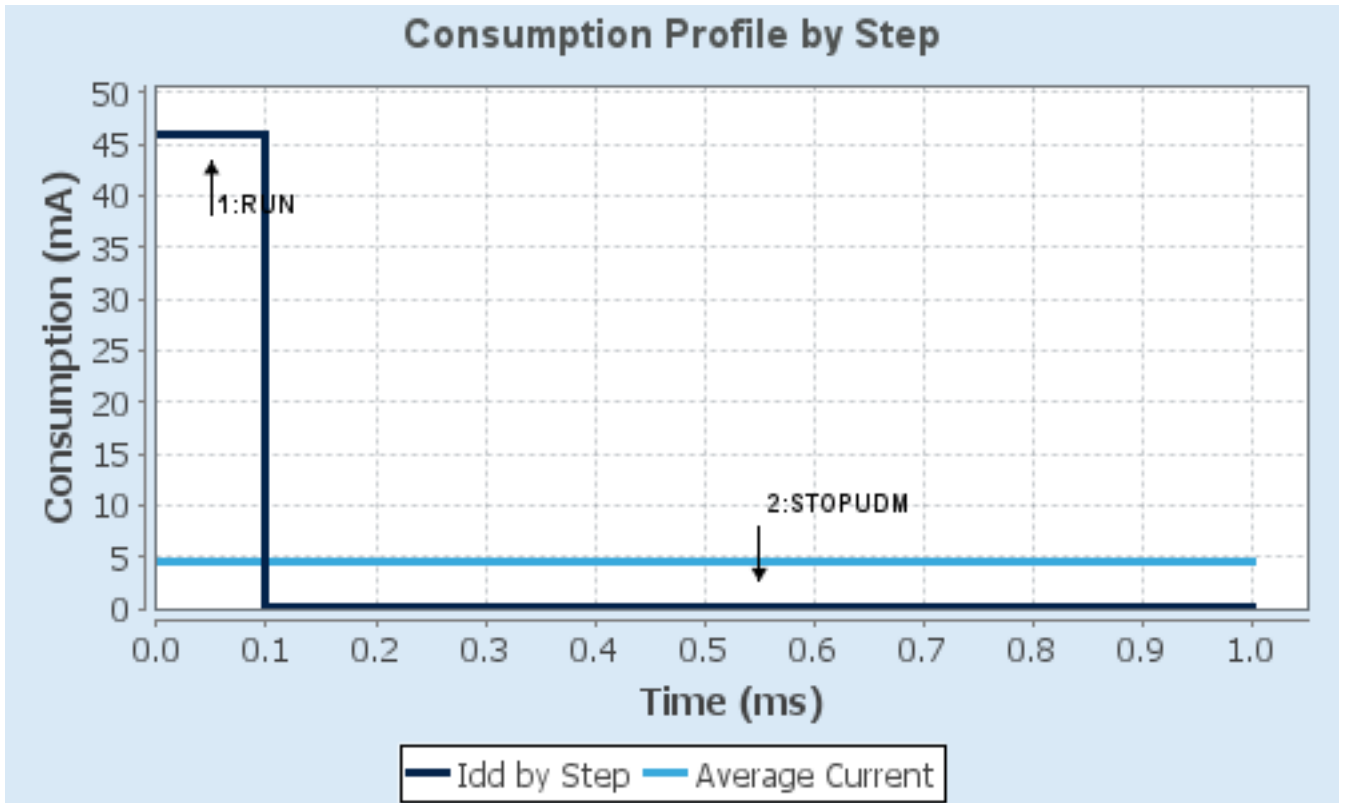
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP_UDM (Under Drive)
Vdd	3.3	3.3
Voltage Source	Battery	Battery
Range	Scale1-High	No Scale
Fetch Type	RAM/FLASH/REGON/ART/P REFETCH	n/a
CPU Frequency	180 MHz	0 Hz
Clock Configuration	HSE PLL	Regulator_LP Flash-PwrDwn
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	46 mA	55 μ A
Duration	0.1 ms	0.9 ms
DMIPS	225.0	0.0
Ta Max	98.02	104.99
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	4.65 mA
Battery Life	1 month	Average DMIPS	225.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	XDIC_TRIM_JIG
Project Folder	C:\working\01_Project\Jig\XD12_JIG\XD12_ES1\FW
Toolchain / IDE	EWARM V8.32
Firmware Package Name and Version	STM32Cube FW_F4 V1.28.0
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	Yes
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	SystemClock_Config	RCC
3	MX_DMA_Init	DMA
4	MX_USART2_UART_Init	USART2
5	MX_TIM12_Init	TIM12
6	MX_SPI2_Init	SPI2
7	MX_TIM5_Init	TIM5
8	MX_SPI1_Init	SPI1
9	MX_ADC1_Init	ADC1
10	MX_TIM8_Init	TIM8
11	MX_TIM1_Init	TIM1

Rank	Function Name	Peripheral Instance Name
12	MX_SPI3_Init	SPI3
13	MX_TIM2_Init	TIM2

3. Peripherals and Middlewares Configuration

3.1. ADC1

mode: IN8

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler PCLK2 divided by 4

Resolution 12 bits (15 ADC Clock cycles)

Data Alignment Right alignment

Scan Conversion Mode Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

End Of Conversion Selection EOC flag at the end of single channel conversion

ADC_Regular_ConversionMode:

Number Of Conversion 1

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel 8

Sampling Time **480 Cycles ***

ADC_Injected_ConversionMode:

Number Of Conversions 0

WatchDog:

Enable Analog WatchDog Mode false

3.2. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

mode: Master Clock Output 2

3.2.1. Parameter Settings:

System Parameters:

VDD voltage (V) 3.3

Instruction Cache Enabled

Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	4 WS (5 CPU cycle)

RCC Parameters:

HSI Calibration Value	16
TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
Power Over Drive	Disabled

3.3. SPI1

Mode: Full-Duplex Master

3.3.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	16 Bits *
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	4 *
Baud Rate	18.4375 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	2 Edge *

Advanced Parameters:

CRC Calculation	Disabled
NSS Signal Type	Software

3.4. SPI2

Mode: Full-Duplex Master

3.4.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	18.4375 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	2 Edge *

Advanced Parameters:

CRC Calculation	Disabled
NSS Signal Type	Software

3.5. SPI3

Mode: Full-Duplex Master

3.5.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	4 *
Baud Rate	9.21875 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSS Signal Type	Software

3.6. SYS

Debug: Serial Wire

Timebase Source: SysTick

3.7. TIM1

Clock Source : Internal Clock

Channel1: PWM Generation CH1

3.7.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	117 *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High

Break And Dead Time management - Output Configuration:

Automatic Output State	Enable *
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.8. TIM2

Clock Source : Internal Clock

Combined Channels: PWM Input on CH2

3.8.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

PWM Input CH2:

Input Trigger	TI2FP2
Slave Mode Controller	Reset Mode

____ Parameters for Channel 1 ____

Polarity Selection

IC Selection

Prescaler Division Ratio

Input Filter (4 bits value)

____ Parameters for Channel 2 ____

Polarity Selection (opposite CH1)

IC Selection

Prescaler Division Ratio

Input Filter (4 bits value)

Falling Edge *

Indirect

No division

0

Rising Edge

Direct

No division

0

3.9. TIM5

mode: Clock Source

Channel1: Input Capture direct mode

3.9.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Enable *

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Input Capture Channel 1:

Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

3.10. TIM8

Clock Source : Internal Clock

Channel2: PWM Generation CH2

3.10.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	179 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	6867 *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High

Break And Dead Time management - Output Configuration:

Automatic Output State	Enable *
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	5 *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.11. TIM12

mode: Clock Source

3.11.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	72 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division

auto-reload preload

Enable *

3.12. USART2

Mode: Asynchronous

3.12.1. Parameter Settings:

Basic Parameters:

Baud Rate	921600 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

* User modified value

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PB0	ADC1_IN8	Analog mode	No pull-up and no pull-down	n/a	XD12_ADO
RCC	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	
	PH0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
	PC9	RCC_MCO_2	Alternate Function Push Pull	No pull-up and no pull-down	Low	MCLK
SPI1	PA5	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA6	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA7	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
SPI2	PC1	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	ADC_SPI_MOSI
	PC2	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	ADC_SPI_MISO
	PB10	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	ADC_SPI_CLK
SPI3	PB2	SPI3_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PC10	SPI3_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PC11	SPI3_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	TMS
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	TCK
TIM1	PA8	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	PWM_OUT
TIM2	PB3	TIM2_CH2	Alternate Function Push Pull	No pull-up and no pull-down		PWM_IN

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
					Very High *	
TIM5	PA0-WKUP	TIM5_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	DOUT_Q4
TIM8	PC7	TIM8_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	VSYN
USART2	PA2	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	USART_TX
	PA3	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	USART_RX
GPIO	PC13	GPIO_Input	Input mode	Pull-up *	n/a	B1
	PC0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	LTC_HIGH_CURRENT
	PC3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	PWM_SWITCH
	PA1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	LTC_LOW_CURRENT
	PA4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	LTC_MID_CURRENT
	PC4	GPIO_EXTI4	External Interrupt Mode with Falling edge trigger detection	Pull-up *	n/a	ADC_DRDY
	PC5	GPIO_Input	Input mode	Pull-up *	n/a	KEY_TRIMMING
	PB1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	ENABLE_SELECT4
	PB12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	ENABLE_SELECT3
	PB13	GPIO_Input	Input mode	Pull-up *	n/a	KEY_POWER
	PB14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	XD12_5_7V
	PB15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	CNT_MR
	PC6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	SPI3_CS
	PC8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	ENABLE_SELECT1
	PA9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	ADC_CS
	PA10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ADC_START
	PA11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED_TRIM

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PA12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	ENABLE_SELECT2
	PC12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	LTC_DISCHARGE
	PB4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	VLED_CTR_9V
	PB5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	XD12_VCC_EN
	PB6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	XC24_VCC_EN
	PB8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ADC_RESET
	PB9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	NSCS

4.2. DMA configuration

DMA request	Stream	Direction	Priority
TIM1_CH1	DMA2_Stream1	Memory To Peripheral	Low
TIM5_CH1	DMA1_Stream2	Peripheral To Memory	Low
TIM2_CH2/CH4	DMA1_Stream6	Peripheral To Memory	Low

TIM1_CH1: DMA2_Stream1 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Half Word
 Memory Data Width: Half Word

TIM5_CH1: DMA1_Stream2 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Word
 Memory Data Width: Word

TIM2_CH2/CH4: DMA1_Stream6 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: **Half Word ***
 Memory Data Width: **Half Word ***

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
EXTI line 4 interrupt	true	0	0
DMA1 stream2 global interrupt	true	0	0
DMA1 stream6 global interrupt	true	0	0
USART2 global interrupt	true	0	0
TIM8 update interrupt and TIM13 global interrupt	true	0	0
DMA2 stream1 global interrupt	true	0	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1, ADC2 and ADC3 interrupts	unused		
TIM1 break interrupt and TIM9 global interrupt	unused		
TIM1 update interrupt and TIM10 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM11 global interrupt	unused		
TIM1 capture compare interrupt	unused		
TIM2 global interrupt	unused		
SPI1 global interrupt	unused		
SPI2 global interrupt	unused		
TIM8 break interrupt and TIM12 global interrupt	unused		
TIM8 trigger and commutation interrupts and TIM14 global interrupt	unused		
TIM8 capture compare interrupt	unused		
TIM5 global interrupt	unused		
SPI3 global interrupt	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
EXTI line 4 interrupt	false	true	true
DMA1 stream2 global interrupt	false	true	true
DMA1 stream6 global interrupt	false	true	true
USART2 global interrupt	false	true	true
TIM8 update interrupt and TIM13 global interrupt	false	true	true
DMA2 stream1 global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware					
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System Core	Analog	Timers	Connectivity	Multimedia	Computing
DMA ✓	ADC1 ✓	TIM1 ✓	SPI1 ✓		
GPIO ✓		TIM2 ✓	SPI2 ✓		
IIVIC ✓		TIM5 ✓	SPI3 ✓		
RCC ✓		TIM8 ✓	USART2 ✓		
SYS ✓		TIM12 ✓			

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32f446_bsdI.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32f446_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32f4-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstmcsuite.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2606-stm32-microcontroller-system-memory-boot-mode-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack-mcus-

and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2945-stm8s-and-stm32-mcus-a-consistent-832bit-product-line-for-painless-migration-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3070-managing-the-driver-enable-signal-for-rs485-and-iolink-communications-with-the-stm32s-usart-stmicroelectronics.pdf

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